

Chapter 1 - The History of Payment Networks

From Cash to Credit Cards: How We Accidentally Invented One of the Greatest Business Models Ever

Why Should Anyone Study Payment Networks?

If I asked you to name the most powerful companies in the world, you'd probably think of Apple, Microsoft, Nvidia, Amazon, or Google.

Very few people would immediately say **Visa** or **Mastercard**.

That's understandable. Unlike Apple, they don't make shiny gadgets. Unlike Netflix, they don't produce TV shows. Unlike Tesla, they don't build cars.

In fact, most people interact with Visa or Mastercard every day without giving them a second thought.

You tap your debit card to buy coffee.

You order shoes online.

You subscribe to Spotify.

You pay your Netflix bill.

Behind the scenes, there's a good chance a payment network quietly moved money between your bank and the merchant in a matter of seconds.

No fanfare. No headlines. Just an invisible piece of infrastructure doing exactly what it was designed to do.

That's what makes payment networks fascinating.

The best infrastructure businesses often become invisible. Electricity works because you don't think about the power grid every time you flip a light switch. The internet works because you don't stop to appreciate the thousands of routers carrying your messages. Payment networks operate in much the same way. Their greatest achievement is making something incredibly complicated feel effortless.

Today, Visa and Mastercard together process **tens of billions of transactions each year**, connecting consumers, merchants, financial institutions, and governments across more than **200 countries and territories**. Their networks support payment volumes measured in the **trillions of U.S. dollars annually**, making them among the most important pieces of financial infrastructure in the global economy. Their role extends far beyond plastic cards—they underpin e-commerce, mobile wallets, recurring subscriptions, and countless digital payment experiences. (Visa and Mastercard annual reports)

From an investor's perspective, payment networks are even more interesting.

These companies don't lend most of the money.

They don't manufacture physical products.

They don't own millions of stores.

Yet they consistently generate extraordinary profit margins, enormous free cash flow, and some of the highest returns on invested capital found in public markets.

That raises an obvious question:

How did two companies that most people barely notice become two of the greatest businesses ever built?

To answer that, we have to travel back to a world where buying something was surprisingly inconvenient.

Section 1 — The World Before Cards

Imagine you're living in 1940.

You want to buy groceries.

Simple enough, right?

Not really.

You first need to make sure you've got enough cash in your wallet. If you don't, you'll have to visit your bank, withdraw money during banking hours, and hope the shopkeeper has enough change.

Now imagine buying something expensive—a refrigerator, a sewing machine, or a plane ticket.

Walking around with a large bundle of cash wasn't just inconvenient. It was risky.

Lose your wallet?

The money was gone.

Get robbed?

Also gone.

Cash had one major advantage: everyone accepted it.

But it also had obvious limitations.

It couldn't travel electronically.

It couldn't be used for online purchases (not that the internet existed yet).

It offered no purchase history, no fraud protection, and no easy way to spend beyond the cash you physically carried.

For centuries, though, it was the best system people had.

Checks: A Step Forward... and Two Steps Slower

To solve some of cash's problems, banks popularized **checks**.

Instead of handing over physical currency, you could write a signed piece of paper instructing your bank to pay someone else.

That was a huge improvement for larger transactions.

People no longer needed to carry large sums of cash.

Businesses could receive payments without transporting bags of money across town.

Sounds great.

Except...

Checks were painfully slow.

A merchant couldn't instantly know whether the account contained enough money.

Banks had to verify signatures.

Documents had to move between financial institutions.

Settlement could take days.

Sometimes even longer.

If you've ever complained because an online payment took five seconds instead of two, imagine waiting an entire week just to know whether someone actually paid you.

Merchants weren't exactly thrilled.

Bank Transfers: Useful, But Not Practical for Everyday Life

Banks also offered transfers between accounts.

Again, this worked reasonably well for certain situations.

Businesses paying suppliers?

Sure.

Governments moving funds?

Absolutely.

Ordinary people buying lunch?

Not so much.

Transfers often required paperwork, bank visits, or manual processing. Cross-border transfers were even worse. Different banking systems, currencies, regulations, and communication standards made international payments slow, expensive, and unpredictable.

Sending money overseas wasn't impossible.

It just required patience and sometimes enough paperwork to make you wonder whether you were applying for citizenship instead.

Commerce Was Growing Faster Than Payments

Here's what I find interesting.

By the middle of the twentieth century, the world was changing rapidly.

People travelled more.

Cities expanded.

International trade accelerated.

Consumer spending increased.

Businesses were becoming larger and more interconnected.

But the payment system hadn't kept pace.

Commerce had entered the modern age.

Payments were still living in the past.

There was a growing mismatch between how quickly people wanted to buy things and how slowly money could move.

And history has a funny habit of rewarding whoever solves a widespread frustration.

Sometimes the greatest business opportunities don't come from inventing something completely new.

They come from fixing something that everyone secretly hates.

The world didn't wake up one morning demanding credit cards.

It simply grew tired of carrying cash, writing checks, and waiting for banks to catch up.

That frustration created one of the biggest commercial opportunities of the twentieth century.

Section 2 — The Birth of Credit Cards

Ironically, the first successful payment card wasn't invented because someone wanted to revolutionize finance.

It was invented because one man forgot his wallet.

According to the widely repeated story, in 1949 businessman **Frank McNamara** was having dinner at New York's Major's Cabin Grill when he realized he couldn't pay the bill. His wife eventually brought the money, but the incident sparked a simple idea:

What if one card could let trusted customers pay later instead of carrying cash?

Whether every detail of the story is perfectly accurate has been debated by historians, but the result certainly wasn't.

In **1950**, McNamara and his partners launched **Diners Club**.

The original card wasn't even made of plastic.

It was cardboard.

It could initially be used at a relatively small number of restaurants, and customers received one monthly bill instead of paying every meal individually.

That might not sound revolutionary today.

Back then, it absolutely was.

For the first time, convenience itself became a product.

People weren't just buying dinner.

They were buying the ability to stop worrying about carrying enough cash.

American Express Enters the Game

The success of Diners Club didn't go unnoticed.

In **1958**, **American Express**, already a trusted name in travel and financial services, launched its own charge card.

Unlike Diners Club, American Express brought something incredibly valuable:

Trust.

People already knew the company.

Merchants already recognized the brand.

That reputation helped American Express expand beyond restaurants into travel, entertainment, hotels, and retail.

The card also became something more than a payment method.

It became a status symbol.

Owning an American Express card quietly signaled that you were a frequent traveler or a successful professional.

Long before premium credit cards came with airport lounges and metal finishes, American Express had figured out that payments weren't only about moving money.

Sometimes they were about identity.

Why Credit Cards Changed Commerce

Most people think credit cards simply replaced cash.

I think they did something much bigger.

They changed **when** money could move.

Before cards, buying something depended almost entirely on whether you had enough cash at that exact moment.

Credit cards separated **the purchase** from **the payment**.

That single change unlocked enormous possibilities.

Consumers gained flexibility.

Merchants attracted more customers.

Banks earned new revenue.

Economic activity became smoother because purchases no longer had to wait until payday.

In hindsight, credit cards weren't just a financial innovation.

They were an infrastructure innovation.

They allowed commerce to move faster than physical money ever could.

And once people experienced that convenience, there was no going back.

What began as a solution to forgotten wallets would eventually become the foundation of a global payment ecosystem handling trillions of dollars every year.

The next challenge wasn't convincing people to use cards.

It was figuring out how to build a payment network large enough for the entire world.

Section 3 — The Birth of Visa

If Diners Club proved that people liked paying with cards, **Visa proved that cards could become a global payment system.**

Those are two very different achievements.

Diners Club and American Express issued their own cards and dealt directly with customers. That worked well while the business remained relatively small.

But by the late 1950s, banks had begun asking a different question:

What if banks—not a single company—could issue cards?

That idea led to one of the most influential financial experiments in history.

BankAmericard

In 1958, **Bank of America** launched **BankAmericard** in Fresno, California.

Unlike earlier charge cards, BankAmericard allowed customers to **revolve credit**, meaning they could carry a balance instead of paying the full amount every month. Today that sounds ordinary. At the time, it was revolutionary.

Bank of America didn't start cautiously.

It mailed tens of thousands of unsolicited cards directly to households.

The experiment was ambitious.

It was also chaotic.

Fraud exploded.

People who had never asked for a credit card suddenly had one.

Defaults increased.

Counterfeit cards appeared.

Banks lost millions.

Many people believed the experiment had failed.

Yet hidden beneath the chaos was a brilliant idea.

Consumers loved the convenience.

Merchants loved larger purchases.

Banks saw enormous long-term potential.

The problem wasn't the idea.

The problem was the system behind it.

Enter Dee Hock

Every great company has a moment where someone doesn't invent the product, but reinvents the organization.

For Visa, that person was **Dee Hock**.

When BankAmericard expanded by licensing its program to banks across the United States, it created a new problem.

Thousands of independent banks wanted to cooperate.

At the same time, they were competitors.

Each wanted control.

None wanted to surrender it.

Traditional corporate structures weren't designed for this.

Dee Hock proposed something unusual.

Instead of one company controlling everyone else, participating banks would jointly govern the network through a decentralized structure.

He later described this philosophy as "**chaordic**"—a blend of **chaos** and **order**.

It sounds like something a management consultant would invent after drinking too much coffee.

Surprisingly, it worked.

Banks shared common rules.

Cards issued by one bank could be accepted by another.

Merchants no longer needed separate agreements with dozens of institutions.

Consumers suddenly had a card that worked almost everywhere.

That simple idea dramatically increased the value of joining the network.

Every new bank attracted more merchants.

More merchants attracted more customers.

More customers attracted even more banks.

Without calling it that, Visa had begun building one of history's strongest **network effects**.

From BankAmericard to Visa

By the mid-1970s, BankAmericard had outgrown its original identity.

The name still reflected one bank.

The business had become global.

In **1976**, BankAmericard officially became **Visa**.

The new name was intentionally simple.

Easy to pronounce.

Easy to translate.

Easy to recognize almost anywhere in the world.

It also carried an important symbolic message.

Visa was no longer merely an American banking product.

It wanted to become **the world's payment network**.

The timing couldn't have been better.

International travel was increasing.

Global trade was expanding.

Consumer spending was growing.

The world needed a payment system that worked across borders.

Visa positioned itself to become exactly that.

Global Expansion

Visa didn't expand by opening thousands of bank branches.

It expanded by convincing more financial institutions to join its network.

Every new issuing bank made Visa more useful for consumers.

Every new merchant accepting Visa made the card more valuable.

Growth became self-reinforcing.

This is one reason Visa scaled so efficiently.

Unlike manufacturers, it didn't need to build factories in every country.

Instead, it expanded through partnerships with local financial institutions while maintaining common network standards.

Today Visa operates in more than **200 countries and territories**, processing billions of transactions annually and connecting consumers, merchants, issuers, acquirers, governments, and fintech companies through a single global network.

My Interpretation

The biggest innovation wasn't the plastic card.

It was the **system**.

People often think Visa became successful because it invented better cards.

I don't think that's true.

The real invention was creating a network where competitors had a reason to cooperate.

Banks competed for customers.

But they cooperated on infrastructure.

That's incredibly rare.

It's similar to every airline agreeing to use the same air traffic control system.

They compete fiercely.

But everyone benefits when the underlying infrastructure works.

Visa understood something many businesses still struggle to grasp:

Owning the network is often more valuable than owning the product.

Section 4 — The Birth of Mastercard

Success rarely goes unanswered.

BankAmericard's rapid expansion worried other banks.

If Bank of America controlled the dominant card system, competing banks risked becoming dependent on a rival.

Naturally, they decided to build their own.

The Interbank Card Association

In **1966**, several California banks formed the **Interbank Card Association (ICA)**.

Its goal wasn't merely to create another credit card.

Its goal was to create a competing payment network that no single bank controlled.

This distinction mattered enormously.

Instead of one dominant institution calling the shots, member banks would collaborate while remaining independent.

The association launched **Master Charge**, which later became **Mastercard** in 1979.

The competition between Visa and Mastercard had officially begun.

Why Mastercard Was Created

At first glance, Mastercard looks like Visa's younger sibling.

The reality is more interesting.

Mastercard wasn't created because consumers wanted another card.

It was created because banks wanted another **network**.

Competition wasn't about plastic.

It was about governance.

Banks wanted bargaining power.

Merchants wanted alternatives.

Consumers benefited because wider acceptance meant greater convenience.

Healthy competition also accelerated innovation.

Both companies invested heavily in fraud prevention, electronic authorization, international acceptance, and technological improvements.

Ironically, neither company could afford to stand still because the other existed.

The Visa–Mastercard Rivalry

For decades, Visa and Mastercard have competed in almost every market.

Banks choose which cards to issue.

Merchants decide which cards to accept.

Consumers decide which card to use.

Yet something unusual happened.

Despite intense competition, both companies became enormously successful.

Normally, fierce rivals destroy each other's profitability.

Visa and Mastercard largely didn't.

Why?

Because they weren't competing in a winner-takes-all market.

The global payments market itself kept expanding.

Cash usage declined.

Electronic commerce increased.

International travel grew.

Online shopping exploded.

The pie became much larger.

Both companies captured enormous slices of it.

Today they remain competitors.

But they're also beneficiaries of the same long-term trends.

Sometimes the biggest opportunity isn't defeating your rival.

It's growing an industry so large that both of you win.

My Interpretation

People love asking,

"Which is better—Visa or Mastercard?"

I think that's the wrong question.

A better question is:

**Why has the industry been able to support
two giants for so long?**

The answer lies in trust.

Payments are one of the few industries where reliability matters more than novelty.

Nobody wants the world's most "innovative" payment network if it fails every third transaction.

Banks prefer stability.

Merchants prefer acceptance.

Consumers prefer convenience.

Once a payment network reaches global scale, replacing it becomes incredibly difficult.

That's why Visa and Mastercard compete aggressively while still benefiting from enormous barriers to entry.

Section 5 — Timeline

Before 1950

Cash, checks, and bank transfers dominate commerce. Payments are slow, local, and largely paper-based.

1950

Diners Club launches the first widely adopted charge card.

**1958**

Bank of America introduces BankAmericard.
American Express launches its charge card.

**1966**

Interbank Card Association (ICA) is formed to compete with BankAmericard.

**1976**

BankAmericard is renamed Visa, reflecting its global ambitions.

**1979**

Master Charge becomes Mastercard.

**1990s**

Electronic authorization becomes standard.
Global card acceptance expands rapidly.



2000s

E-commerce accelerates.

Online payments become mainstream.



2010s

Contactless payments, smartphones, and digital wallets transform consumer behavior.



2026

Visa and Mastercard operate as critical financial infrastructure,
powering trillions of dollars in payment volume each year across more than 200 countries and territories.

Conclusion

It's easy to look at Visa and Mastercard today and assume their success was inevitable.

It wasn't.

Their rise was built on decades of experimentation, competition, mistakes, and gradual improvements in how money moves.

The story also teaches an important investing lesson.

The companies that change the world aren't always the ones with the flashiest products.

Sometimes they're the ones solving incredibly boring problems—better than anyone else.

Nobody wakes up excited to think about payment authorization or clearing systems.

Yet every online purchase, every airline ticket, every hotel booking, every subscription, and millions of everyday transactions depend on those systems working flawlessly.

History also reveals something deeper.

Visa and Mastercard didn't become extraordinary because they invented the credit card.

Credit cards already existed.

They became extraordinary because they built **the infrastructure that allowed millions of banks, merchants, and consumers to trust one another at scale.**

In my opinion, that's the real beginning of the investment story.

Before you can understand why Visa and Mastercard earn exceptional margins, generate enormous free cash flow, and possess durable competitive advantages, you first have to understand **how they became the rails on which modern commerce runs.**

The history isn't just background information.

It's the foundation for understanding everything that comes next.

Chapter 2 - How Payment Networks Work

Introduction

Every time someone taps a card at a coffee shop, the entire transaction feels almost magical.

Tap.

Approved.

Coffee secured.

Life goes on.

Most people assume the money instantly moves from their bank account to the merchant's account.

It doesn't.

In reality, that simple "Approved" message is the result of multiple financial institutions talking to each other in a matter of seconds. Banks verify identities, payment networks route information, fraud systems scan for suspicious activity, and settlement processes ensure everyone gets paid.

What's remarkable isn't that this happens occasionally.

It's that it happens **billions of times every single day**, often in less time than it takes you to decide whether you want fries with your burger.

Understanding this process is essential because Visa and Mastercard don't actually move money the way many people think they do. Their greatest contribution

isn't lending money or holding customer deposits, it's making sure every participant in the payment ecosystem can communicate with one another efficiently and securely.

1. The Four Participants

Every card transaction involves four key players. Remove even one of them, and the entire payment process falls apart.

1. Cardholder

The cardholder is the person making the purchase.

Whether it's buying groceries, booking a flight, or paying for an online subscription, the transaction begins with the customer.

The cardholder receives a credit or debit card from an issuing bank and agrees to follow the terms attached to it, such as repayment schedules, credit limits, or account balances.

Although the customer initiates the transaction, they rarely think about everything happening behind the scenes.

And honestly, that's exactly how a good payment system should work.

Good infrastructure is almost invisible.

2. Merchant

The merchant is the business accepting the payment.

This could be:

- A neighborhood grocery store
- An online retailer
- A luxury hotel
- A multinational airline

Merchants accept cards because customers increasingly expect fast, convenient payment options.

Cash may still exist, but consumers often choose convenience over counting change.

In return for accepting card payments, merchants pay processing fees to participate in the payment ecosystem.

Think of those fees as the cost of reaching millions of customers who prefer paying electronically.

3. Issuing Bank

The issuing bank is the financial institution that provides the card to the customer.

Examples include banks like JPMorgan Chase, HDFC Bank, SBI, or Bank of America.

The issuing bank is responsible for:

- Issuing the card
- Verifying customer identity
- Approving or declining transactions
- Managing credit limits
- Sending monthly statements
- Collecting repayments

If you're using a credit card, it's usually the issuing bank—not Visa or Mastercard—that lends you the money.

This is one of the biggest misconceptions people have about payment networks.

4. Acquiring Bank

If issuing banks represent customers, acquiring banks represent merchants.

The acquiring bank provides businesses with the ability to accept card payments.

Its responsibilities include:

- Processing merchant transactions
- Receiving payment requests
- Communicating with payment networks
- Settling funds with merchants

Without acquiring banks, merchants would have no standardized way to accept electronic payments at scale.

My Interpretation

I like to think of these four participants as members of a relay race.

The customer starts with the baton.

The merchant passes it forward.

Banks exchange information.

The payment network coordinates the handoff.

If one runner drops the baton, nobody wins.

Payments aren't a one-company business.

They're a team sport.

2. Visa & Mastercard's Role

Here's the biggest surprise for many people:

Visa and Mastercard usually don't lend money.

That responsibility belongs to issuing banks.

Instead, Visa and Mastercard focus on something arguably even more important.

They operate the network that allows banks to communicate with each other quickly, securely, and consistently.

Imagine thousands of banks around the world trying to exchange payment information without common standards.

It would be chaos.

Visa and Mastercard solve that problem.

They provide:

- Transaction routing
- Authorization systems
- Security standards
- Fraud detection tools
- Settlement infrastructure
- Global interoperability

Because they connect thousands of financial institutions through one standardized network, a card issued in New York can be used at a café in Tokyo or a bookstore in London with remarkably little friction.

My Interpretation

People often call Visa a "credit card company."

I don't think that's the best description.

Visa is closer to the **internet of money**.

The internet doesn't create your emails.

It simply makes sure they reach the right destination.

Visa and Mastercard do something very similar for financial transactions.

They don't care whether you're buying pizza or plane tickets.

Their job is simply to make sure the payment gets where it needs to go—quickly, accurately, and securely.

3. How One Transaction Works

A card payment may take only a second from the customer's perspective, but several steps occur behind the scenes.

Customer taps card



Merchant



Acquiring Bank



Visa / Mastercard



Issuing Bank



Approval or Decline



Settlement

Let's walk through each step.

Step 1 – Customer Taps the Card

The customer inserts, taps, or swipes their card to initiate a purchase.

The merchant's payment terminal captures the transaction details.

Step 2 – Merchant Sends the Request

The merchant forwards the payment request to its acquiring bank.

The acquiring bank packages the transaction and prepares it for routing.

Step 3 – Visa or Mastercard Routes the Request

The payment network identifies which issuing bank owns the card.

It securely forwards the authorization request to that bank.

Notice something important:

Visa doesn't decide whether you have enough money.

It simply knows where to send the request.

Step 4 – Issuing Bank Makes the Decision

The issuing bank checks several factors, including:

- Available credit or account balance
- Card validity
- Fraud indicators
- Spending patterns
- Security verification

The bank then either approves or declines the transaction.

Step 5 – Approval Returns

The decision travels back through the same path:

Issuing Bank → Visa/Mastercard → Acquiring Bank → Merchant.

If approved, the customer sees one of the happiest words in retail:

Approved.

If declined...

Well, suddenly checking your bank app becomes everyone's favorite activity.

Step 6 – Settlement

Approval doesn't immediately transfer the money.

Actual movement of funds typically occurs later through a settlement process.

The issuing bank transfers funds to the acquiring bank, which then deposits the appropriate amount into the merchant's account after deducting agreed processing fees.

Settlement ensures everyone involved receives what they're owed.

My Interpretation

Authorization is like making a dinner reservation.

Settlement is actually paying the restaurant after you've eaten.

People often confuse the two because they happen so seamlessly.

4. Revenue Sources

At first glance, Visa and Mastercard appear to earn money every time someone uses a credit card.

That's true—but the story is more nuanced.

Their revenue comes from several different sources.

1. Assessment Fees

Assessment fees are charged based on the total payment volume flowing across the network.

As consumer spending grows, payment volume increases, and these fees generally rise alongside it.

This creates a business that naturally benefits from long-term economic growth.

2. Data Processing Fees

Every transaction requires:

- Authorization
- Routing
- Security checks
- Clearing
- Settlement support

Visa and Mastercard earn fees for processing these transactions through their networks.

As electronic payments continue replacing cash, transaction volumes increase, strengthening this revenue stream.

3. Cross-Border Fees

International transactions are typically more complex than domestic ones.

They involve:

- Currency conversion
- Additional security checks
- Regulatory compliance
- Greater processing complexity

Visa and Mastercard therefore earn higher fees on many cross-border transactions.

This is one reason both companies benefit from trends such as international tourism, global trade, and cross-border e-commerce.

4. Value-Added Services

Over time, payment networks have expanded beyond simple transaction processing.

Today they offer services including:

- Fraud prevention
- Tokenization
- Cybersecurity
- Data analytics
- Risk management
- Consulting services
- Open banking solutions

These businesses often generate attractive margins because they build on the existing network rather than requiring entirely new infrastructure.

My Framework: The Toll Road Business

The easiest way I think about Visa and Mastercard is this:

Imagine the world's busiest toll road.

The companies don't manufacture the cars.

They don't own the fuel.

They don't decide where people travel.

They simply built the road that everyone wants to use.

Every time another car drives across it, they collect a small toll.

One toll doesn't sound exciting.

Billions of tolls every year become one of the greatest business models ever created.

Conclusion

The beauty of payment networks lies in their simplicity from the user's perspective and their sophistication behind the scenes. Every transaction depends on four participants working together, with Visa and Mastercard acting as the trusted communication layer that keeps the entire system running smoothly.

Perhaps the most surprising lesson is that these companies don't need to lend money or own customer deposits to build extraordinary businesses. Instead, they earn revenue by providing the infrastructure that enables billions of transactions to occur reliably every year.

In many ways, Visa and Mastercard resemble the digital highways of modern commerce. The more people shop, travel, and conduct business electronically, the more valuable those highways become. That's what makes payment networks so fascinating from an investment

perspective—they don't need to predict what consumers will buy next. They simply need to make sure that, whatever people choose to buy, the payment gets from one side to the other.

Chapter 3 - Visa vs Mastercard vs American Express

Three Great Businesses. Three Very Different Businesses.

1. Introduction

If there were a Hall of Fame for payment companies, Visa, Mastercard, and American Express would all have permanent seats.

They're constantly compared because they operate in the same broad industry. Consumers use their cards to buy coffee, book flights, pay bills, and occasionally convince themselves that buying another pair of sneakers is an "investment." From the outside, they look almost identical.

They're not.

Comparing these companies only by looking at their cards is like comparing Amazon, FedEx, and UPS because all three deliver packages. The end result may look similar, but what happens behind the scenes is completely different.

Visa and Mastercard are primarily **payment networks**. They build and maintain the financial highways that move money between banks, merchants, and consumers. They don't usually issue cards, and they don't usually lend money.

American Express takes a very different approach.

Instead of simply owning the highway, it owns many of the cars driving on it too. It issues cards, lends directly to customers, processes payments, manages merchant relationships, and operates its own network. That gives it more control over the customer experience—and more opportunities to earn revenue—but it also exposes the company to credit risk that Visa and Mastercard largely avoid.

This distinction is the reason investors continue debating which company is "better."

They're not choosing between three versions of the same business.

They're choosing between **three different economic engines**.

My Framework: The Three Layers of a Payment Company

I like to think about these companies using a simple analogy.

Visa

Builds the roads.

The more traffic on those roads, the more money Visa makes.

It doesn't care whether you're driving a Toyota or a Ferrari.

Mastercard

Builds another set of roads.

The strategy is similar, but Mastercard has invested more aggressively in services surrounding those roads—cybersecurity, fraud prevention, open banking, consulting, and data analytics.

American Express

Owns the road.

Owns many of the cars.

Knows who's driving.

Knows where they're spending.

Sometimes finances the fuel.

That's a much more involved business.

It also means more moving parts.

2. Business Model Comparison

Feature	Visa	Mastercard	American Express
Primary Business Model	Open-loop payment network	Open-loop payment network	Closed-loop payment network + issuer
Issues Cards Directly?	No (partner banks issue cards)	No (partner banks issue cards)	Yes (primarily issues its own cards)
Lends Money?	No	No	Yes
Network Type	Open Loop	Open Loop	Closed Loop
Primary Revenue Sources	Service fees, data processing, cross-border fees	Service fees, data processing, cross-border fees, value-added services	Merchant discount revenue, interest income, annual fees, network services
Carries Credit Risk?	Minimal	Minimal	Significant
Primary Customer Focus	Everyone	Everyone	Premium consumers & businesses
Relationship with Banks	Partner	Partner	Often competitor and partner simultaneously

Why This Matters

Visa and Mastercard are essentially toll-road businesses.

Every time money crosses their network, they collect a small fee.

Whether the cardholder pays the bank back tomorrow or next month isn't really Visa's problem.

American Express doesn't have that luxury.

If customers don't repay their balances, Amex feels it directly.

That additional risk is also why American Express earns interest income that Visa and Mastercard simply don't.

More risk.

More reward.

3. Financial Comparison

(Approximate FY2024 figures)

Metric	Visa	Master card	American Express
Revenue	~\$36.5B	~\$28.2B	~\$66.0B

Net Income	~\$20.0B	~\$13.8B	~\$10.1B
Operating Margin	~66%	~58%	~19%
Return on Equity (ROE)	~49%*	~180%*	~34%
Free Cash Flow	~\$19B	~\$13B	~\$9B
Market Capitalization	~\$700B+	~\$500B+	~\$220B+

*ROE for Visa and Mastercard is unusually high because of share repurchases and relatively low shareholder equity. It should be interpreted alongside other profitability measures.

Why Are The Numbers So Different?

Looking only at revenue might lead someone to conclude that American Express is the strongest business because it reports the highest revenue.

That would be the wrong conclusion.

Revenue isn't created equally.

American Express records interest income because it lends money.

Visa and Mastercard generally don't.

Think of it this way.

Suppose two restaurants each earn \$1 million.

One owns the building.

The other rents it.

One may report higher expenses and lower margins despite generating the same sales.

Business models matter.

The same logic applies here.

Visa and Mastercard convert an astonishing percentage of revenue into profit because they're essentially running software-powered financial infrastructure.

American Express runs a lending business on top of its payment network.

Loans require reserves.

Reserves reduce margins.

My Margin Rule

Whenever I compare companies, I ask one question:

"How much of every extra dollar of revenue actually becomes profit?"

Visa and Mastercard answer that question with a smile.

American Express answers with a loan portfolio.

Neither is wrong.

They're simply different businesses.

4. Competitive Advantages

Network Effects

Winner: Visa

Every additional cardholder makes Visa more valuable to merchants.

Every additional merchant makes Visa more valuable to consumers.

That's the textbook definition of a network effect.

Mastercard enjoys the same advantage, though on a slightly smaller scale.

American Express benefits from network effects too, but because of its premium positioning and smaller merchant base, those effects aren't quite as powerful globally.

Brand

Winner: American Express

Visa and Mastercard have incredibly recognizable brands.

American Express has something rarer.

Prestige.

Many consumers actively want an Amex card—not simply because it works, but because of what it represents.

It's one of the few financial brands that people occasionally brag about owning.

Nobody has ever posted,

"Just got approved for my Visa Debit Card 🔥"

Merchant Acceptance

Winner: Visa

Visa enjoys the broadest acceptance worldwide.

Mastercard follows closely behind.

American Express has improved dramatically over the last decade but still trails because of historically higher merchant fees.

Convenience matters.

If your card isn't accepted, its rewards don't help much.

Customer Base

Winner: American Express

Visa and Mastercard serve billions of cardholders.

American Express serves fewer people.

But they're often the people spending the most.

For payment companies, one customer who spends \$50,000 annually can be worth more than five customers spending \$5,000 each.

Quality sometimes beats quantity.

Global Presence

Winner: Visa

Visa remains the largest payment network globally.

Mastercard isn't far behind.

American Express maintains strong international operations but remains more concentrated in premium markets.

5. Which Business Is Better?

This question depends on what you mean by "better."

The best business isn't always the best investment.

And the best investment isn't always the biggest company.

So instead of picking one winner, let's compare them across the traits that matter most.

Business Quality

Winner: Visa

Visa earns extraordinary returns while taking remarkably little credit risk.

Its business is simple, scalable, and incredibly difficult to disrupt.

Risk

Winner: Mastercard

Like Visa, Mastercard avoids lending risk. It also has a diversified mix of value-added services that reduce dependence on transaction fees.

Both are low-risk businesses by industry standards, but Mastercard's continued expansion into cybersecurity, fraud prevention, and consulting adds another layer of resilience.

Growth

Winner: Mastercard

Visa is already enormous.

Mastercard starts from a slightly smaller base, giving it a bit more room to expand internationally and grow adjacent businesses.

Profitability

Winner: Visa

Margins that regularly exceed 60% are almost unfair.

Visa turns payment volume into profit with remarkable efficiency.

It's the financial equivalent of owning a toll booth on the busiest highway in the world.

Long-Term Durability

Winner: Visa

Consumer spending changes.

Technology changes.

Cards become phones.

Phones become watches.

Maybe someday we'll all pay by blinking at a checkout camera.

Through all of those changes, money still needs a trusted network to move securely.

Visa's infrastructure remains deeply embedded in the global financial system.

Replacing it would be like trying to replace the internet during lunch break.

6. Final Verdict

Every one of these companies deserves respect.

Each has produced outstanding returns for shareholders over long periods.

But if I had to rank them strictly as businesses—not considering short-term valuation—my list would look like this.

1st — Visa

Visa combines the strongest network effects, exceptional profitability, global acceptance, minimal credit risk, and unmatched scale.

It doesn't need customers to borrow money.

It simply benefits every time they spend it.

That's a beautiful business model.

2nd — Mastercard

Mastercard shares many of Visa's strengths while standing out through innovation, cybersecurity, data services, and international growth opportunities.

If Visa is the industry leader, Mastercard is the competitor that refuses to stay comfortably in second place.

The gap between them is smaller than many people assume.

3rd — American Express

Third place sounds disappointing.

It shouldn't.

American Express is still one of the highest-quality financial businesses in the world.

Its premium brand, affluent customer base, and integrated ecosystem create advantages that few competitors can match.

The reason it ranks below Visa and Mastercard is structural.

Lending introduces credit risk.

Credit risk makes earnings more cyclical.

Great business.

Slightly more complicated economics.

Final Thoughts

Comparing Visa, Mastercard, and American Express isn't really about deciding which logo belongs in your wallet. It's about deciding which economic model you want in your portfolio.

Visa monetizes **scale**.

Mastercard monetizes **scale plus innovation**.

American Express monetizes **relationships**.

If I had to summarize them in one sentence each:

- **Visa** is the highway.
- **Mastercard** is the smarter highway with better traffic analytics.
- **American Express** is the private club with valet parking.

All three can get you to the destination.

They just charge for the journey in very different ways.

Chapter 4 - Why Payment Networks Are Incredible

The Quiet Toll Booths of the Global Economy

Introduction

Some businesses sell products.

Some sell services.

Payment networks sell something much stranger.

They sell **trust**.

Every time someone taps a card in Tokyo, buys coffee in New York, books a hotel in Paris, or pays tuition in Mumbai, an invisible system quietly asks one question:

"Can we move this money safely in under two seconds?"

Most people never think about it.

That's exactly why it's such a great business.

The best infrastructure businesses are almost invisible. You don't notice electricity until the lights go out. You don't think about the internet until Wi-Fi disappears. And you rarely think about Visa or Mastercard until the card terminal flashes **"Transaction Declined."**

Payment networks have quietly become some of the highest-quality businesses ever built. They process trillions of dollars every year without manufacturing

products, carrying inventory, or worrying whether consumers actually repay their credit card bills.

That combination is exceptionally rare.

This chapter explores why.

1. Network Effects

Every great business has an advantage.

Payment networks have an advantage that compounds itself.

Economists call this a **network effect**.

I like to call it **the party problem**.

Imagine hosting a party.

Nobody wants to be the first guest.

But once fifty people arrive, everyone suddenly wants an invitation.

Payment networks behave the same way.

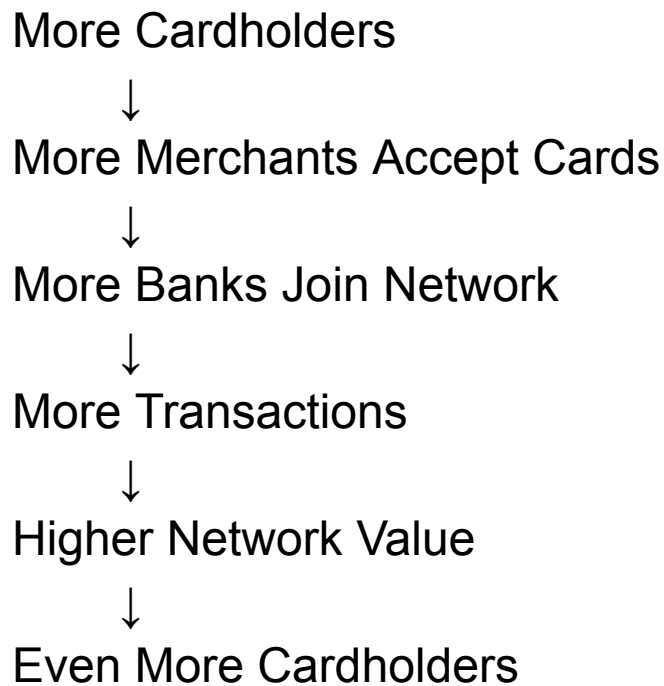
More cardholders attract more merchants.

More merchants attract more cardholders.

More transactions attract more banks.

More banks issue more cards.

The cycle feeds itself.



Notice something interesting.

Visa doesn't need to convince every consumer individually.

It mostly needs to keep strengthening the network.

The network does much of the selling on its own.

That's why network effects are so difficult to compete against.

A new payment company doesn't just need better technology.

It needs millions of consumers, millions of merchants, thousands of banks, global fraud systems, regulatory approvals, and decades of trust.

That's not competition.

That's mountain climbing with flip-flops.

My Framework: The Snowball Advantage

I think network effects are easiest to visualize as a snowball.

At first, it's tiny.

You push it.

Nothing much happens.

Then it starts rolling.

Eventually the snowball becomes so large that gravity does most of the work.

Visa reached that stage years ago.

Today, the network itself creates momentum.

Management still matters.

Innovation still matters.

But scale has become a competitive weapon.

2. Asset-Light Business Model

One of the first things new investors notice is how little Visa and Mastercard actually own.

No factories.

No warehouses.

No fleets of trucks.

No giant inventory sitting on shelves.

Instead, they own software, payment infrastructure, intellectual property, data centers, security systems, and one incredibly valuable asset:

The network itself.

That's the beauty of an asset-light business.

Growth doesn't require building thousands of physical locations.

When transaction volume doubles, Visa doesn't need twice as many buildings.

It mostly needs more computing power and continued investment in technology.

Compare that with airlines.

When demand grows, airlines need:

- More aircraft
- More pilots
- More fuel
- More maintenance
- More airports
- More headaches

Visa mostly needs more transactions.

One business scales with concrete and steel.

The other scales with code.

Investors tend to prefer the second.

My Rule

Whenever I see revenue growing much faster than physical assets, I pay attention.

That's usually a sign that the business can scale efficiently.

Payment networks are one of the clearest examples of that principle.

3. Pricing Power

Pricing power is one of those phrases investors love using.

Sometimes without knowing what it actually means.

It isn't the ability to raise prices.

Almost every company can raise prices once.

Pricing power is the ability to raise prices **without significantly reducing demand**.

Visa and Mastercard possess exactly that.

Merchants don't choose payment networks because they're cheap.

They choose them because customers expect them.

Imagine a retailer deciding to stop accepting Visa.

Technically possible.

Commercially painful.

If customers walk away because their preferred payment method isn't accepted, saving a fraction of a percent in fees suddenly looks much less attractive.

That's why pricing discussions often happen quietly.

No dramatic announcements.

No flashy marketing.

Just small increases spread across billions of transactions.

Tiny percentages.

Massive dollars.

The "Coffee Test"

Suppose Visa earns just a few extra cents from millions of daily purchases.

Nobody notices.

Consumers still buy coffee.

Merchants still make sales.

Banks still issue cards.

But multiplied across hundreds of billions of transactions, those tiny increases become enormous.

Sometimes the most profitable businesses don't charge more.

They charge **slightly more** an astonishing number of times.

4. Switching Costs

Switching costs are often misunderstood.

People assume they mean contracts.

They usually don't.

The strongest switching costs are invisible.

Think about everything connected to your primary credit card:

- Streaming subscriptions
- Insurance payments
- Utility bills
- Airline rewards
- Hotel loyalty programs
- Online shopping accounts
- Business expenses
- Mobile wallets

Changing cards isn't impossible.

It's just annoying.

Now imagine you're a bank.

Replacing an established payment network means rebuilding fraud systems, merchant relationships, payment processing, customer support, compliance procedures, and international acceptance.

Suddenly, staying put looks like the cheaper option.

Good businesses make customers happy.

Great businesses make leaving feel inconvenient.

Payment networks quietly achieve both.

5. Capital Allocation

One of the most underrated reasons payment networks have created extraordinary shareholder returns is what they do **after** earning money.

Many companies generate cash.

Fewer know what to do with it.

Visa and Mastercard typically allocate capital through four primary channels:

- Reinvesting in technology and security
- Strategic acquisitions
- Share repurchases
- Dividends

Notice what's missing.

Gigantic factories.

Massive inventory.

Capital-intensive expansion projects.

Because the business doesn't require enormous ongoing physical investment, much of its cash becomes available for shareholders.

That's a wonderful problem to have.

My Capital Allocation Rule

I judge management less by how they make money...

and more by what they do after they make it.

Anyone can celebrate record earnings.

The real test begins the following morning.

6. High Margins

One of the easiest ways to identify an exceptional business is to ask:

"How much of every extra dollar actually stays with the company?"

For Visa and Mastercard, the answer is:

A surprisingly large amount.

Operating margins above 50% are rare.

Above 60%?

Almost unfair.

Why?

Because the cost of processing one additional transaction is tiny compared to the value generated.

Once the infrastructure exists, adding another payment is remarkably inexpensive.

Imagine building a bridge.

The first car is incredibly expensive because you had to build the bridge.

The millionth car costs almost nothing.

Payment networks operate similarly.

The infrastructure requires investment.

After that, scale becomes extraordinarily profitable.

My Observation

High margins alone don't make a company great.

But **high margins combined with recurring demand** are one of my favorite combinations.

Payment networks have both.

7. Buffett's Investment Thesis

Warren Buffett didn't invest heavily in payment businesses because cards are exciting.

He invested because the economics are.

His investment in American Express stretches back decades, surviving recessions, market crashes, inflation, and changing consumer behavior.

Why?

Because Buffett wasn't buying plastic cards.

He was buying:

- A trusted brand
- Recurring transaction volume
- Customer loyalty
- Pricing power
- Strong returns on capital

Those qualities tend to age well.

Interestingly, Buffett never needed consumers to spend more every single year.

He simply needed more commerce to move through durable businesses.

As economies grow, payment networks often grow alongside them.

Sometimes investing isn't about predicting the future.

It's about owning the infrastructure that benefits regardless of which company sells the next popular product.

My Take

If Buffett likes businesses that resemble castles protected by moats, then payment networks don't just have moats.

They have drawbridges, walls, security cameras, and probably a dragon or two.

The moat isn't one feature.

It's the combination of network effects, trust, regulation, brand, technology, and decades of relationships.

8. Long-Term Growth Drivers

The best businesses don't rely on a single growth story.

Payment networks benefit from several powerful trends happening simultaneously.

The Decline of Cash

Every purchase that shifts from cash to digital payments expands the opportunity.

The world doesn't have to spend more.

It simply has to spend differently.

E-Commerce

Online shopping cannot function at scale without secure payment infrastructure.

As digital commerce grows, payment networks become increasingly essential.

Mobile Wallets

Apple Pay.

Google Pay.

Samsung Wallet.

People often assume these threaten Visa and Mastercard.

Ironically, many of those wallets still use Visa and Mastercard behind the scenes.

The wallet changed.

The rails stayed the same.

Cross-Border Payments

International travel, global trade, and online marketplaces continue increasing cross-border transaction volume.

These transactions often generate higher fees than domestic payments.

More passports.

More profits.

Emerging Markets

Billions of people are gradually entering the formal financial system.

As banking access, smartphones, and internet penetration improve, electronic payments continue replacing cash.

That runway is measured in decades, not quarters.

Value-Added Services

Payment companies increasingly earn revenue from:

- Fraud prevention
- Cybersecurity
- Tokenization
- Identity verification
- Data analytics
- Open banking
- Real-time payments

They're becoming financial technology platforms, not just transaction processors.

That's an important evolution.

Final Thoughts

Payment networks are remarkable because they don't need to predict which phone you'll buy, which airline you'll fly, or which streaming service you'll binge this weekend.

They simply benefit when money moves.

That's an unusually powerful position.

Visa and Mastercard don't have to win every consumer decision. They just need to remain the trusted rails beneath those decisions.

The longer I study these businesses, the less I think of them as "credit card companies."

They're closer to digital infrastructure.

Electricity powers cities.

Roads move cars.

Payment networks move commerce.

And history has shown that owning infrastructure—especially infrastructure people forget exists—is often one of the quietest ways to build extraordinary long-term wealth.

As investors, we're often tempted to chase the company inventing the next shiny gadget. Sometimes that's worthwhile. But sometimes the better investment is the

company collecting a tiny toll every time that gadget is purchased.

It's less glamorous.

It's also a wonderfully boring way to make money.

Chapter 5 - The Future of Payment Networks

Will the Rails of Global Commerce Change—or Simply Get Faster?

Introduction

Every few years, someone declares that Visa and Mastercard are finished.

Sometimes it's cryptocurrencies.

Sometimes it's fintech.

Sometimes it's Buy Now, Pay Later.

Recently, it's been artificial intelligence, stablecoins, central bank digital currencies, and real-time payment systems.

The headlines rarely change.

Only the villain does.

If you read enough financial news, you'd think Visa has been five years away from extinction for the last twenty years.

Yet something interesting keeps happening.

Transaction volumes continue rising.

Revenue continues growing.

Free cash flow keeps compounding.

So who's right?

The innovators claiming they'll replace traditional payment networks?

Or the incumbents quietly processing trillions of dollars every year?

As with most questions in investing, the answer isn't black and white.

Technology absolutely changes payment systems.

But history suggests it often changes **the interface**, not necessarily **the infrastructure**.

The credit card disappeared from your wallet.

It moved into your phone.

The payment network underneath barely noticed.

This chapter explores the technologies most likely to shape the future of payments—not by asking *what is new*, but by asking a more useful question:

Who actually captures the value when payments evolve?

That distinction matters far more than the technology itself.

My Framework: Payments Have Three Layers

Whenever a new payment technology appears, I ask three questions.

Layer 1 — Interface

How does the customer pay?

Card.

Phone.

Watch.

QR code.

Voice.

Biometric scan.

Layer 2 — Network

Who actually moves the money?

Visa?

Mastercard?

Bank transfer?

Real-time payment system?

Layer 3 — Settlement

Where does the money ultimately come from?

Bank account.

Credit line.

Digital wallet.

Stablecoin.

Central bank money.

Most innovation happens at Layer One.

Most profits still live in Layers Two and Three.

That simple distinction explains why many "payment revolutions" end up helping incumbents instead of replacing them.

1. Digital Wallets

If someone had predicted fifteen years ago that millions of people would happily pay for coffee by waving a rectangle of glass in front of a machine, they probably would've sounded slightly unwell.

Today we call it Tuesday.

Digital wallets have fundamentally changed consumer behavior.

Apple Pay.

Google Pay.

Samsung Wallet.

PayPal.

They've made payments faster, more secure, and more convenient.

Many investors initially viewed them as existential threats to Visa and Mastercard.

The logic seemed straightforward.

If consumers stop using cards, card networks lose relevance.

Except...

that's not what happened.

Most digital wallets still rely on Visa and Mastercard for authorization, tokenization, fraud prevention, and settlement.

The card disappeared.

The rails remained.

My Observation

People often confuse **brand visibility** with **economic value**.

Apple Pay feels like the payment.

In reality, it often functions more like a beautifully designed front door.

The house behind it still belongs to Visa or Mastercard.

Technology frequently changes what consumers see while leaving the underlying infrastructure largely intact.

2. UPI

India's Unified Payments Interface (UPI) has become one of the most important payment innovations of the past decade.

It enables instant bank-to-bank transfers at extraordinary scale, often with minimal or no transaction fees for consumers.

From street vendors to luxury retailers, QR codes have become nearly universal.

UPI demonstrates that governments and regulators can successfully build national payment infrastructure.

Naturally, this raises an important question.

If countries develop their own payment systems...

do global card networks become less valuable?

The answer depends on geography.

UPI works exceptionally well inside India.

Cross-border commerce is a different challenge.

Global tourism.

International e-commerce.

Foreign business payments.

These still require international acceptance, settlement infrastructure, fraud management, and regulatory coordination.

Building a domestic payment system is difficult.

Building a trusted global one is substantially harder.

My Framework: Local Champion vs Global Highway

UPI is an excellent city metro.

Visa is an international airline.

Both move people.

They solve different transportation problems.

That's why I don't necessarily view them as direct substitutes.

They're optimized for different journeys.

3. Real-Time Payments

Consumers increasingly expect money to move instantly.

Waiting two or three business days feels almost prehistoric in a world where movies stream instantly and food arrives before your motivation to cook disappears.

Real-time payment systems aim to eliminate that delay.

Countries around the world are expanding instant payment infrastructure.

Examples include:

- FedNow (United States)
- Faster Payments (United Kingdom)
- PIX (Brazil)
- UPI (India)

These systems reduce settlement times and improve efficiency.

Does that threaten Visa?

Partially.

Certain bank-to-bank transfers may bypass traditional card networks.

But real-time payments don't eliminate the need for:

- Fraud prevention
- Identity verification
- Dispute resolution
- Global acceptance
- Cross-border interoperability

Speed solves one problem.

Commerce requires many others.

My View

The future isn't simply "faster payments."

It's **trusted instant payments**.

Those are two different products.

4. Central Bank Digital Currencies (CBDCs)

Central Bank Digital Currencies have generated enormous discussion among economists and policymakers.

A CBDC is essentially a digital version of sovereign currency issued directly by a central bank.

Unlike cryptocurrencies, CBDCs represent official government money.

Supporters argue they could improve financial inclusion, payment efficiency, and monetary policy implementation.

Critics worry about privacy, cybersecurity, and excessive government oversight.

From an investment perspective, one question dominates.

Could CBDCs eliminate traditional payment networks?

Personally, I think that's unlikely.

Creating digital money isn't the difficult part.

Building global acceptance, merchant infrastructure, fraud systems, consumer protections, and commercial relationships is.

Money and payment networks are related.

They're not identical.

My Framework

Currency answers:

"What are we paying with?"

Payment networks answer:

"How does it move safely?"

Changing the first doesn't automatically replace the second.

5. Stablecoins & Crypto

Few technologies have attracted as much excitement—and confusion—as cryptocurrencies.

Some enthusiasts envision a future where blockchain replaces every existing payment network.

Others dismiss the entire industry outright.

Reality, as usual, lives somewhere in the middle.

Stablecoins offer genuine advantages in certain situations:

- Cross-border transfers
- Programmable payments
- Continuous settlement
- Lower costs in specific corridors

These are meaningful innovations.

But widespread retail payments require more than efficient technology.

They require consumer trust, regulatory clarity, merchant acceptance, dispute mechanisms, and seamless user experiences.

Those are areas where traditional payment networks have spent decades building expertise.

Interestingly, Visa and Mastercard haven't ignored crypto.

They've partnered with exchanges, experimented with blockchain settlement, supported stablecoin initiatives, and invested in tokenization.

Incumbents often adapt faster than critics expect.

My Observation

History suggests established infrastructure rarely disappears overnight.

Railroads weren't destroyed by trucks.

They specialized.

Banks weren't eliminated by online banking.

They digitized.

Payment networks may evolve in exactly the same way.

6. Buy Now, Pay Later (BNPL)

Buy Now, Pay Later changed how many consumers finance purchases.

Companies like Klarna, Affirm, and Afterpay popularized installment payments integrated directly into online checkout experiences.

For merchants, BNPL often increases conversion rates.

For consumers, it reduces immediate financial friction.

The danger, of course, is that financial friction occasionally exists for a reason.

Buying expensive headphones feels much easier when the payment is divided into four smaller installments.

Mathematics didn't change.

Psychology did.

Initially, many investors believed BNPL would weaken credit card usage.

Instead, something more nuanced happened.

Banks, Visa, Mastercard, and American Express responded by launching their own installment products or partnering with BNPL providers.

The incumbents adapted.

Again.

My Rule

Whenever someone says:

"This new technology will destroy the industry."

I ask a different question.

"What if the industry simply adopts it?"

That possibility receives surprisingly little attention.

7. AI in Payments

Artificial intelligence will almost certainly reshape payments.

Not because AI processes transactions faster.

Because it makes payment systems smarter.

Areas likely to benefit include:

Fraud Detection

AI can recognize suspicious transaction patterns almost instantly.

The better the models become, the fewer fraudulent transactions reach consumers.

Personalized Financial Services

Banks and payment companies can increasingly offer customized credit products, spending insights, budgeting tools, and financial recommendations.

Customer Support

AI assistants reduce operational costs while improving response times.

Risk Assessment

Machine learning improves underwriting, credit decisions, merchant monitoring, and anomaly detection.

Operational Efficiency

From cybersecurity to compliance monitoring, AI automates tasks previously requiring large teams of analysts.

My Framework: AI Is A Multiplier

AI rarely replaces an economic moat.

It magnifies one.

A company with excellent data becomes better.

A company with poor data becomes a faster version of itself.

That's an important distinction.

Visa processes enormous volumes of payment data.

That gives AI unusually rich information to learn from.

In many industries, data creates AI advantages.

In payments, AI also strengthens the value of existing data.

That's a reinforcing cycle.

8. Will Visa & Mastercard Stay Dominant?

This is the question that quietly sits behind every discussion about payment innovation.

My answer is:

Probably—but not because nothing changes.

Because they keep changing too.

Investors sometimes imagine disruption as a boxing match.

One company enters.

One company leaves.

Business history usually looks more like urban development.

New buildings appear.

Old buildings get renovated.

Roads expand.

Infrastructure adapts.

Visa and Mastercard have repeatedly shown an ability to absorb technological change rather than resist it.

Magnetic stripes became chip cards.

Chip cards became contactless payments.

Contactless became mobile wallets.

Now they're investing in tokenization, AI, real-time payments, open banking, and stablecoin settlement.

That's not the behavior of businesses waiting to be disrupted.

It's the behavior of businesses attempting to participate in every future payment method.

Could their economics change?

Absolutely.

Could competition increase?

Certainly.

Will regulation become more aggressive?

Almost inevitably.

But replacing a global payment network involves much more than writing better software.

It requires convincing governments, banks, merchants, regulators, consumers, and financial institutions across hundreds of countries to trust an entirely new system.

That's an extraordinarily high hurdle.

My Investment Framework: Follow the Rails, Not the Wallet

When evaluating payment businesses over the next decade, I think investors should focus less on the app consumers open and more on the infrastructure that quietly powers the transaction.

Ask questions like:

- Who authorizes the payment?
- Who verifies identity?
- Who prevents fraud?
- Who manages disputes?
- Who handles cross-border settlement?

- Who benefits regardless of whether the customer taps a card, scans a QR code, or uses a smartwatch?

Technology changes quickly.

Infrastructure changes slowly.

That's often where durable investment opportunities live.

Final Thoughts

The future of payments won't be defined by one winner replacing everyone else.

It will be defined by layers.

Consumers will gain more ways to pay.

Merchants will gain more ways to accept payments.

Governments will build faster domestic systems.

Fintech companies will design smoother user experiences.

Artificial intelligence will make transactions safer.

Stablecoins will improve specific use cases.

Real-time payments will reduce delays.

And through much of that change, someone will still need to authenticate identities, route transactions, enforce standards, resolve disputes, detect fraud, and

connect thousands of financial institutions across borders.

That role doesn't disappear simply because payments become more modern.

If anything, it becomes more valuable.

Perhaps the biggest lesson from studying payment networks is this:

The companies most likely to survive technological revolutions aren't always the ones inventing every new feature.

They're often the ones quietly becoming essential to every feature everyone else invents.

Or, to put it another way:

People love talking about the next shiny payment app.

Investors should spend at least as much time studying the rails underneath it.

Because history has a habit of rewarding the businesses that collect a tiny toll every time the future arrives.

Chapter 6 - Risks & Challenges Facing Payment

Every Great Business Has Enemies. The Best Ones Simply Learn to Live With Them.

Introduction

It's easy to fall in love with great businesses.

High margins.

Network effects.

Global brands.

Rivers of free cash flow.

Visa and Mastercard have all of those.

That's precisely why investors need to spend time studying what could go wrong.

Charlie Munger once said that inversion is a powerful way of thinking. Instead of asking, *"Why will this company succeed?"* ask:

"What could cause this business to become less valuable?"

That's the purpose of this chapter.

Not to argue that payment networks are in decline—they're not—but to understand the forces constantly trying to weaken their position.

Interestingly, many of the biggest "threats" don't actually want to replace Visa or Mastercard.

They simply want to own a larger piece of the payment value chain.

Sometimes they succeed.

Sometimes they end up making Visa and Mastercard even stronger.

As investors, separating genuine disruption from noisy headlines is one of the most valuable skills we can develop.

My Framework: The Four Categories of Risk

I find it helpful to divide payment network risks into four buckets.

1. Regulatory Risk

Governments changing the rules.

2. Technological Risk

New payment methods replacing old ones.

3. Competitive Risk

Someone builds a better alternative.

4. Operational Risk

The network itself stops working—or loses trust.

Every threat discussed in this chapter fits somewhere within these four categories.

1. Regulatory Pressure

If there's one thing governments dislike, it's when private companies become too important.

Visa and Mastercard process trillions of dollars every year.

That naturally attracts regulatory attention.

Governments care about payment systems because payments aren't just commerce.

They're infrastructure.

Over the past two decades, regulators across the United States, European Union, Australia, India, and several other jurisdictions have examined issues such as:

- Interchange fees
- Merchant fees
- Competition
- Anti-trust concerns
- Cross-border pricing
- Consumer protection
- Data privacy

The central concern is straightforward.

If a payment network becomes indispensable, could it begin charging excessive fees?

That's why regulators periodically intervene.

The European Union, for example, has capped certain interchange fees.

Australia has introduced similar reforms.

These actions reduce profitability in specific areas but rarely threaten the existence of the networks themselves.

My Observation

Regulation usually doesn't destroy infrastructure businesses.

It changes the economics around the edges.

Electric utilities remain valuable despite regulation.

Railroads survived regulation.

Banks survived regulation.

Payment networks probably will too.

The important question isn't:

"Will regulation happen?"

It almost certainly will.

The better question is:

"Can the business remain highly profitable afterward?"

History suggests yes.

2. UPI & Government Payment Rails

India's Unified Payments Interface (UPI) has become one of the world's most successful real-time payment systems.

Consumers can transfer money instantly using mobile apps.

Merchants often pay little or no transaction fee.

Adoption has exploded.

Naturally, investors began asking:

"If every country builds something like UPI, what happens to Visa and Mastercard?"

It's a reasonable concern.

But it requires separating **domestic payments** from **global payments**.

UPI excels at domestic bank-to-bank transfers.

Visa and Mastercard specialize in something much broader:

- Cross-border payments
- Credit transactions

- International merchant acceptance
- Fraud protection
- Global interoperability

Building a domestic payment rail is difficult.

Building one that works seamlessly across hundreds of countries is an entirely different challenge.

My Framework: Roads vs Highways

I think of UPI as a brilliantly designed city road.

Visa is an international highway system.

If you're traveling across town, the road may be better.

If you're crossing continents, you'll probably still want the highway.

Different infrastructure solves different problems.

3. Stablecoins

If you spend enough time on finance Twitter—or whatever it's called by the time you're reading this—you'll eventually encounter someone claiming stablecoins will eliminate every bank, payment processor, and central bank before lunch.

Reality tends to be less dramatic.

Stablecoins solve genuine problems.

They enable:

- Faster settlement
- Lower-cost cross-border transfers
- Programmable payments
- Continuous transactions

Those are meaningful innovations.

However, consumers don't choose payment systems based only on settlement speed.

They also care about:

- Fraud protection
- Chargebacks
- Ease of use
- Merchant acceptance
- Consumer trust
- Regulation

Stablecoins still face challenges in many of those areas.

Interestingly, Visa and Mastercard aren't ignoring the technology.

Both have invested in blockchain initiatives, stablecoin settlement pilots, and partnerships with digital asset firms.

That's an important pattern.

Incumbents frequently adopt useful innovations rather than resisting them.

My Observation

Technology doesn't automatically eliminate incumbents.

Sometimes incumbents simply buy the technology—or integrate it.

History is full of "disruptions" that became product features.

4. Central Bank Digital Currencies (CBDCs)

Central Bank Digital Currencies represent digital versions of sovereign currencies issued directly by central banks.

Unlike cryptocurrencies, CBDCs remain government liabilities.

Supporters believe they could:

- Reduce payment costs
- Improve financial inclusion
- Modernize monetary systems
- Improve payment efficiency

Critics worry about:

- Privacy
- Government surveillance
- Cybersecurity
- Banking system disruption

Would CBDCs eliminate Visa?

Probably not.

A CBDC changes **what** money looks like.

Visa manages **how** money moves.

Those are different functions.

Imagine replacing paper cash with digital cash.

Consumers would still require:

- Authentication
- Fraud detection
- Merchant systems
- Global acceptance
- Transaction routing
- Customer support

Digital money still needs digital infrastructure.

My Framework

CBDCs may replace paper.

They don't automatically replace payment networks.

Money and movement are separate businesses.

5. Fintech Competition

Few words excite investors more than **fintech**.

Startups promise:

- Lower fees
- Better user experiences
- Faster payments
- Smarter budgeting
- Instant lending

Many deliver exactly that.

Companies like Block, Stripe, PayPal, Adyen, Klarna, and Revolut have transformed parts of the financial ecosystem.

But here's something interesting.

Many fintech companies still rely on Visa and Mastercard underneath their products.

Stripe often routes card payments through the existing networks.

Apple Pay frequently uses tokenized Visa and Mastercard credentials.

Even digital wallets commonly depend on traditional payment rails.

The interface changed.

The plumbing often didn't.

My Observation

Consumers usually interact with fintech brands.

Fintech companies often interact with Visa.

That's a remarkably comfortable position to occupy.

Where Fintech Really Competes

Fintech isn't always attacking Visa directly.

It's often competing against:

- Banks
- Branches
- Paper forms
- Slow software
- Terrible user interfaces

Those are much easier targets.

6. AI & Fraud Prevention

Artificial intelligence creates both opportunity and risk.

Fraudsters now possess better tools.

Fortunately, so do payment networks.

Modern fraud detection increasingly relies on machine learning models capable of identifying unusual transaction patterns in milliseconds.

Future AI systems will likely improve:

- Identity verification
- Fraud detection

- Credit scoring
- Risk management
- Customer service
- Compliance monitoring

The interesting question isn't whether AI matters.

It certainly does.

The question is who benefits most.

Large payment networks possess enormous datasets.

That data improves AI.

Better AI reduces fraud.

Lower fraud strengthens trust.

Greater trust increases network usage.

The cycle reinforces itself.

My Framework: AI Multiplies Existing Advantages

AI doesn't magically create a competitive moat.

It usually amplifies the moat that already exists.

Companies with the best data generally improve fastest.

Visa processes hundreds of billions of transactions annually.

That's an extraordinary training dataset.

7. Cybersecurity

Every digital payment ultimately depends on one fragile asset.

Trust.

Consumers tap cards because they believe the transaction will be secure.

Merchants accept cards because they trust settlement.

Banks issue cards because they trust the network.

A major cybersecurity failure could damage that trust.

Consequently, payment companies invest billions in:

- Encryption
- Tokenization
- Multi-factor authentication
- Fraud monitoring
- Cloud security
- Threat intelligence
- AI-powered detection

Cybersecurity isn't simply an IT expense.

It's core infrastructure.

Without it, the entire business model weakens.

My Observation

Consumers rarely notice excellent cybersecurity.

That's exactly how it's supposed to work.

The best security systems are invisible.

The worst ones become newspaper headlines.

8. Future Outlook

So...

after all these risks...

what does the future actually look like?

Personally, I think payment networks become **less visible but more essential**.

Consumers will increasingly pay using:

- Phones
- Watches
- Cars
- Smart glasses
- Voice assistants
- Biometric authentication

Eventually, perhaps devices we haven't invented yet.

The interface will continue evolving.

The infrastructure will evolve alongside it.

Visa and Mastercard are already investing in:

- AI
- Tokenization
- Open banking
- Stablecoin settlement
- Digital identity
- Cybersecurity
- Real-time payments

That doesn't guarantee success.

But it demonstrates something important.

They're not standing still.

My Investment Framework: Follow the Infrastructure

Whenever I evaluate new payment technologies, I ask four questions.

Who owns the customer?

Who owns the data?

Who owns the infrastructure?

Who earns money regardless of which app consumers prefer?

Those questions often lead back to surprisingly familiar companies.

Final Thoughts

Every generation believes it's witnessing the technology that will permanently change payments.

Checks were supposed to replace cash.

Cards replaced checks.

Digital wallets replaced cards.

Stablecoins may transform parts of the system.

AI will almost certainly reshape fraud detection.

Government payment rails will continue expanding.

Each innovation changes something.

Few replace everything.

The biggest mistake investors make is assuming disruption always destroys incumbents.

Sometimes it strengthens them.

Visa and Mastercard have survived the internet, e-commerce, smartphones, contactless payments, mobile wallets, fintech, and crypto—not because they resisted change, but because they absorbed it.

That's an underrated competitive advantage.

The future of payments probably won't belong to one winner.

It will belong to an ecosystem.

Banks.

Fintechs.

Governments.

Payment networks.

Digital wallets.

AI systems.

Each will own a different layer of the stack.

For investors, the challenge isn't predicting which shiny app people will use five years from now.

It's identifying which companies continue collecting a small toll no matter **how** people choose to pay.

History suggests those businesses tend to age remarkably well.

After all, fashions change.

Infrastructure usually invoices everyone anyway.

Chapter 7 - Future of Payment Networks

The Interface Will Change. The Economics Underneath May Not.

Introduction

The payment industry has spent the last twenty years being declared obsolete.

First, physical cards were supposed to disappear.

Then digital wallets were expected to replace card networks.

Then cryptocurrencies were going to remove banks from the equation.

Now stablecoins, central bank digital currencies, artificial intelligence, open banking, and embedded finance are all competing for the honour of becoming the next great payments revolution.

Some of these technologies will matter enormously.

Most will not eliminate the existing system as neatly as their promotional material suggests.

Payments are often discussed as though they are a single product. In reality, a payment involves several separate functions: identity verification, authorization, fraud detection, transaction routing, currency conversion, settlement, consumer protection, and

dispute resolution. A new technology may improve one of these functions without replacing the rest.

That distinction is central to this chapter.

The future of payments will probably not be defined by one new system defeating every older one. It will be defined by a more complicated structure in which banks, fintech companies, digital wallets, stablecoin issuers, governments, merchants, and global payment networks each control different layers.

The customer may see a cleaner checkout button.

Behind that button, half the financial system may still be arguing over who gets paid.

The investment question is therefore not simply:

Which payment method will become popular?

A better question is:

Which businesses will continue earning attractive returns regardless of how the customer chooses to pay?

My Framework: The Four Layers of Future Payments

To understand where the industry is heading, I divide the payment system into four layers.

Layer	Core Function	Likely Participants
Interface	How the customer initiates payment	Wallets, merchant apps, smart devices
Intelligence	Fraud detection, identity, credit and personalization	AI systems, banks, payment networks, fintechs
Movement	Routing value between parties	Card networks, account-to-account rails, stablecoins
Settlement	Final transfer of money	Commercial banks, central banks, tokenized money

Most visible innovation occurs in the interface.

Most economic power lies deeper in the stack.

A new wallet may become fashionable quickly. A settlement network requires trust, regulation, liquidity, security, and cooperation across thousands of institutions. One can be launched after a successful product meeting. The other usually requires several governments and enough lawyers to populate a small country.

7.1 AI in Payments

Artificial intelligence will influence nearly every stage of the payment process, but its greatest impact will probably be less glamorous than the headlines suggest.

AI is unlikely to replace money.

It will make the systems surrounding money more efficient.

The most immediate application is fraud prevention. Payment companies already rely on large datasets to identify unusual behaviour, and machine-learning systems can examine patterns far faster than human analysts. Future models should become better at distinguishing genuine transactions from fraud without blocking legitimate customers.

That balance matters.

A system that approves everything loses money.

A system that rejects everything technically prevents fraud, but it also prevents commerce. Even the most cautious bank eventually notices that this is not an ideal business model.

AI is also likely to improve:

- identity verification;
- merchant risk assessment;
- credit underwriting;

- transaction monitoring;
- regulatory compliance;
- customer support;
- personalized offers;
- payment routing.

The long-term advantage may belong to companies possessing the largest and most diverse transaction datasets. AI models improve through data, but not all data is equally valuable. Payment data is especially useful because it captures real economic behaviour rather than stated preferences.

A consumer may claim to be financially disciplined.

Their transaction history occasionally provides a more literary interpretation.

Original Analysis: AI as a Moat Multiplier

AI should not be treated as a separate competitive moat in every case. It is better understood as a **moat multiplier**.

A company with extensive data, deep customer relationships and strong distribution can use AI to improve an already powerful business. A smaller competitor may have equally impressive algorithms but lack the transaction history, trust and scale required to deploy them effectively.

This creates a reinforcing cycle:

More transactions → more data → better risk models → lower fraud → greater trust → more transactions

For Visa, Mastercard, American Express, large banks and major payment processors, AI may strengthen the incumbent advantage rather than destroy it.

The risk is that AI also improves the tools available to fraudsters. Synthetic identities, automated scams and increasingly convincing impersonation could raise the cost of maintaining trust. The future will therefore resemble an arms race in which both sides become more capable.

The winners will not merely possess the smartest model. They will possess the data, infrastructure and financial resources to update it continuously.

7.2 Stablecoins

Stablecoins are digital tokens designed to maintain a stable value, usually by linking themselves to a traditional currency.

Their strongest investment case is not that consumers desperately want to purchase groceries using blockchain terminology. Their strongest case is that traditional settlement can be slow, expensive and restricted by banking hours.

Stablecoins may improve:

- cross-border transfers;
- treasury management;
- settlement between financial institutions;
- programmable payments;
- payments in countries with unstable currencies;
- internet-native commerce.

They can move continuously across digital networks and potentially reduce the number of intermediaries involved in certain transactions.

That is a genuine economic advantage.

However, moving value is only one part of a payment. Stablecoins do not automatically provide consumer protection, refunds, fraud guarantees, identity standards or merchant support. These services may still need to be provided by banks, networks or new specialized companies.

Stablecoins as a New Settlement Layer

My view is that stablecoins are more likely to become a **settlement technology** than a complete replacement for payment networks.

Consumers may still pay through familiar wallets. Merchants may still use existing checkout systems. Payment companies may still manage authorization and fraud. The stablecoin could operate quietly underneath, moving value between institutions more efficiently.

In that scenario, stablecoins would not eliminate the payment stack.

They would renovate one floor.

This could still place pressure on traditional cross-border fees, particularly where existing transfers are expensive or slow. Companies earning unusually high fees from inefficient international payment corridors are more vulnerable than networks earning small amounts across enormous transaction volumes.

The main risks remain regulatory uncertainty, reserve quality, liquidity and concentration. A payment asset is only useful while users trust that it will remain stable and redeemable. A stablecoin that suddenly becomes philosophically flexible about the meaning of “stable” will not enjoy a long career.

7.3 Central Bank Digital Currencies

Central bank digital currencies, or CBDCs, are digital forms of sovereign money issued or supported by central banks.

The concept attracts governments for several reasons. CBDCs could potentially improve payment efficiency, expand financial inclusion, reduce cash-handling costs and provide public infrastructure for digital transactions.

Yet the design questions are substantial.

Would consumers hold CBDCs directly with the central bank?

Would commercial banks distribute them?

How much privacy would users retain?

Could payments be programmed or restricted?

What happens to bank deposits if households can hold central-bank money directly?

These are not small technical details. They determine whether a CBDC becomes a modest payment upgrade or a fundamental redesign of the banking system.

CBDCs Will Not Automatically Replace Networks

A digital currency provides a new form of money.

It does not necessarily provide a complete commercial payment system.

Retail commerce still requires merchant acceptance, fraud controls, consumer interfaces, dispute handling and international interoperability. Central banks may issue the money while private companies provide the services around it.

This creates a possible partnership model:

- central banks provide the digital asset;
- commercial banks manage customer relationships;
- technology firms build interfaces;

- payment networks connect domestic systems internationally.

The greatest threat to traditional networks would emerge if several large economies developed interoperable CBDCs that enabled cheap international payments without relying on existing card infrastructure.

That scenario is possible, but coordination would be difficult. Governments do not always agree on trade, regulation or data standards. Expecting them to build one frictionless global payment system may be optimistic. They occasionally struggle to agree on the shape of a conference table.

7.4 Cross-Border Payments

Cross-border payments remain one of the most attractive and contested areas of finance.

Domestic payments have become faster and cheaper in many countries, but international transfers often remain complicated. A single transaction may involve correspondent banks, currency conversion, compliance checks, settlement delays and several layers of fees.

This creates opportunities for:

- stablecoin-based settlement;
- real-time payment links;
- specialized fintech companies;
- multi-currency wallets;

- improved bank infrastructure;
- global payment networks.

The main difficulty is not simply moving information. It is moving regulated value across jurisdictions with different currencies, legal systems and compliance standards.

Cross-border payments are expensive partly because the system is inefficient, but also because the problem itself is genuinely difficult.

The Trust Premium

I describe the excess profitability available in international payments as the **trust premium**.

A company is not paid merely for transferring money. It is paid for ensuring that the money arrives safely, legally, in the correct currency and with somebody available when things go wrong.

Technology should compress some fees over time. However, the businesses capable of combining scale, compliance, liquidity and trust may continue earning strong returns even if individual transaction prices fall.

Visa and Mastercard remain well positioned because they already connect financial institutions globally. Fintech companies may take market share in particular corridors, while stablecoins could improve settlement efficiency. The likely outcome is not one winner but a

more competitive system with lower prices and greater transaction volume.

For investors, falling unit fees are not automatically disastrous if total payment volume expands faster.

A smaller toll on a much busier road can still be excellent business.

7.5 Embedded Finance

Embedded finance refers to financial services being integrated directly into non-financial products and platforms.

Instead of visiting a bank or separate payment provider, customers encounter finance inside the activity they are already performing.

Examples include:

- payments built into retail apps;
- insurance offered during checkout;
- merchant financing within commerce platforms;
- banking services inside business software;
- credit offered at the moment of purchase.

The advantage is convenience. Financial friction is removed from the customer journey.

For businesses, embedded finance can increase customer retention, improve conversion rates and create additional revenue streams. For payment networks and

processors, it expands the number of places where transactions can occur.

The Disappearing Payment

The most important feature of embedded finance is that the payment itself becomes less visible.

Customers do not actively choose a payment system every time they use a ride-hailing app, order food or renew a software subscription. The transaction occurs quietly within the service.

This changes competition.

Brand recognition at the consumer level may become less important, while relationships with platforms, merchants and software providers become more valuable.

Payment companies that offer reliable developer tools, application interfaces, fraud services and global acceptance may benefit even if their brands become almost invisible to the customer.

In payments, invisibility can be a sign of success.

Nobody compliments plumbing when it works. The industry only becomes a topic of conversation when something leaks.

7.6 Open Banking

Open banking allows customers to share financial data or initiate payments through authorized third-party providers, usually with the customer's consent.

Its purpose is to reduce the control traditional banks hold over financial information and encourage competition in products such as payments, lending, budgeting and account management.

Open banking could weaken card networks in certain account-to-account transactions. A merchant may prefer a direct bank payment if it costs less than accepting a card.

However, cards provide benefits that direct transfers may not automatically reproduce:

- credit;
- rewards;
- fraud protection;
- chargebacks;
- international acceptance;
- familiar customer experience.

That means open banking is more likely to take share where these benefits matter less, particularly recurring bills, lower-risk transfers and certain large purchases.

Data Portability Is Not Customer Loyalty

Open banking makes it easier for financial data to move.

It does not guarantee that customers will move with it.

Traditional banks still possess trust, deposits and distribution. Fintech companies may offer better interfaces but struggle with profitability or regulation. Payment networks can respond by providing their own open-banking services and identity tools.

The competitive result will depend on execution rather than ideology.

The company with the most open system will not automatically win. The winner will be the one that combines openness with security, reliability and a business model that eventually produces cash. The last requirement is occasionally treated as an optional feature in fintech.

7.7 Future Competitive Landscape

The future payment system will probably be more fragmented at the interface and more concentrated in infrastructure.

Consumers will use a growing variety of wallets, devices and checkout systems. Yet many of those interfaces may depend on a smaller group of companies for cloud infrastructure, identity, fraud prevention, transaction routing and settlement.

I expect the industry to divide into five broad groups.

1. Global Networks

Visa and Mastercard will continue connecting banks, merchants and consumers while expanding into identity, cybersecurity, data and account-to-account payments.

2. Closed-Loop Ecosystems

Companies such as American Express and large digital platforms will control more of the customer relationship, data and economics within their own systems.

3. Domestic Public Rails

Government-backed real-time systems will dominate many local transfers, particularly where regulators prioritize low-cost payments.

4. Fintech Interfaces

Fintech firms will compete through user experience, specialized products and embedded distribution.

5. Tokenized Settlement Networks

Stablecoins, tokenized deposits and possibly CBDCs will modernize parts of the settlement layer.

These groups will compete, but they will also depend on one another.

The future is unlikely to be a clean war between “old finance” and “new finance.” It will be an uncomfortable partnership in which each side repeatedly announces that it is disrupting the other before signing another commercial agreement.

My Future Payments Framework: Complement, Compress or Replace

When evaluating any new payment technology, I place it into one of three categories.

Category	Meaning	Example Impact
Complement	Strengthens existing networks	AI fraud detection, embedded payments
Compress	Reduces fees or profitability in part of the system	Open banking, real-time domestic rails
Replace	Removes the incumbent from the transaction	Direct settlement through an alternative network

Most innovations belong in the first two categories.

Very few achieve full replacement.

This matters because markets often value new technologies as though replacement is inevitable. In practice, integration is usually more common. Existing

networks possess distribution, regulatory relationships and trust. New entrants possess speed, design and technical flexibility.

The strongest outcome often combines both.

7.8 Investment Outlook

My long-term view on global payment networks remains positive, but not blindly so.

Visa and Mastercard still possess exceptional advantages:

- global acceptance;
- powerful two-sided networks;
- trusted brands;
- capital-light economics;
- high incremental margins;
- strong cash generation;
- deep banking relationships.

Their greatest strength is not that cards will remain unchanged forever. It is that these companies have repeatedly adapted their networks to new interfaces.

Plastic cards may become less visible.

The credentials, security systems and network relationships behind them may remain highly valuable.

Bull Case

In the bullish scenario, electronic payments continue taking share from cash, global commerce expands, cross-border travel grows, and payment networks successfully add new services in fraud prevention, identity, open banking and tokenized settlement.

Under this outcome, Visa and Mastercard become broader commerce infrastructure companies rather than narrowly defined card networks.

Base Case

In the base case, domestic payment rails and open banking take share in certain categories, while regulatory pressure limits some fees. However, transaction growth, international payments and value-added services offset much of that pressure.

The networks grow more slowly but remain highly profitable.

Bear Case

The bearish scenario would involve several threats arriving together:

- domestic account-to-account payments gaining widespread merchant adoption;
- global stablecoin settlement reducing cross-border economics;
- aggressive regulation lowering network fees;

- major cybersecurity failures damaging trust;
- weaker bargaining power against banks and large platforms.

None of these is impossible. The real risk lies in several occurring simultaneously.

My Investment Rating

Factor	Assessment
Industry Growth	Strong
Incumbent Moats	Exceptional
Regulatory Risk	Meaningful
Technological Risk	Manageable but rising
Pricing Power	Strong, though not unlimited
Long-Term Investment Appeal	High at sensible valuations

My overall outlook for the leading global payment networks is **8.8/10**.

The industry deserves a high rating because digital transactions should continue expanding and the major networks remain deeply embedded in global commerce. The score stops below ten because regulation, alternative rails and valuation still matter. Even the finest

toll road becomes a questionable investment if purchased at the price of the entire city.

Final Thoughts

The future of payments will be faster, more invisible and more intelligent.

Artificial intelligence will improve fraud detection.

Stablecoins may modernize settlement.

CBDCs could change the form of sovereign money.

Open banking will increase competition.

Embedded finance will place payments inside everyday products.

Cross-border transactions should become cheaper and more efficient.

Yet the core requirements of a payment system will remain remarkably familiar.

People must trust it.

Merchants must accept it.

Institutions must connect to it.

Regulators must tolerate it.

And somebody must handle the transaction when everything goes wrong at 2:13 a.m. on a public holiday.

That final requirement is where many elegant theories meet the less elegant reality of finance.

The leading payment networks are unlikely to dominate forever simply because they dominate today. Their future depends on whether they can turn new technologies into additional layers of their existing infrastructure.

So far, that has been their most underrated skill.

They do not need to invent every new payment method.

They need to remain useful whenever somebody invents one.

For long-term investors, that may be the most durable thesis of all: the interface will keep changing, but the companies trusted to secure, connect and settle global commerce may continue collecting their share long after the latest payment trend has stopped appearing in investor presentations.

Chapter 8 - Conclusion & Investment Takeaways

The history of payments is usually told as a history of technology.

Cash became cheques. Cheques became cards. Cards became contactless payments. Contactless payments moved into phones, watches, applications, and increasingly invisible digital systems.

That version is correct, but incomplete.

The more important story is the evolution of **trust infrastructure**.

A successful payment system must do more than move money. It must confirm that the buyer is legitimate, assure the merchant that payment will arrive, connect different financial institutions, manage fraud, establish common technical standards, and resolve the occasional disaster when somebody buys airline tickets from a website last updated during the Renaissance.

Visa and Mastercard became extraordinary businesses because they solved these problems without needing to own most of the money being transferred. They created global networks that sit between consumers, merchants, banks, processors, fintech companies, and increasingly digital platforms. Their role is small in relation to the total value of commerce, yet essential to making that commerce work.

That combination—taking a tiny fee from an enormous and growing flow of transactions—is one of the cleanest business models ever created.

The central conclusion of this project is therefore not that cards will remain unchanged forever. They will not. The conclusion is that Visa and Mastercard built something much more durable than plastic cards.

They built the operating rails of global consumer commerce.

8.1 Key Findings

Several findings emerged repeatedly throughout this research.

Payment Networks Are Infrastructure Businesses

Visa and Mastercard are often described as credit-card companies. That description is convenient but misleading.

They generally do not make most of their money by lending directly to consumers. Issuing banks usually provide the credit, hold the customer relationship, and absorb the lending risk. Visa and Mastercard primarily provide the network connecting the different parties involved in a transaction.

This distinction explains much of their economic quality.

A traditional lender must evaluate borrowers, fund loans, reserve against losses, and survive credit cycles. A payment network earns revenue from transaction activity while carrying far less direct credit exposure.

The network does not need to predict whether every consumer will repay.

It needs the transaction to travel safely from one institution to another.

That is an unusually attractive position in the financial value chain.

Network Effects Are the Core Moat

The value of a payment network rises as participation expands.

Consumers prefer cards accepted by many merchants.

Merchants prefer networks used by many consumers.

Banks prefer networks with broad customer demand and merchant acceptance.

Processors and fintech companies integrate with networks that already provide global reach.

Each participant attracts the others.

This creates a self-reinforcing system that becomes more difficult to challenge as it grows.

A competitor does not merely need superior payment software. It must persuade consumers, merchants, issuing banks, acquiring banks, regulators, processors, and technology providers to support a new system at roughly the same time.

That is less like launching an application and more like organizing a global dinner where every guest refuses to arrive until everyone else is already seated.

Scale Improves the Economics

Once the infrastructure exists, additional transactions can be processed at relatively low incremental cost.

This produces several attractive characteristics:

- high operating margins;
- strong free-cash-flow conversion;
- low physical capital requirements;
- significant operating leverage;
- high returns on invested capital.

The business is not completely costless. Payment networks must invest heavily in cybersecurity, fraud prevention, technology, compliance, tokenization, and global operations. But those costs do not generally rise in direct proportion to payment volume.

Revenue can therefore grow faster than the underlying expense base.

Scale does not merely make Visa and Mastercard larger.

It makes them economically stronger.

Trust Is as Important as Technology

Technology is necessary, but it is not the entire moat.

Payments require institutional trust.

Banks must trust the network's standards.

Merchants must trust settlement.

Consumers must trust security and dispute protection.

Regulators must trust the system's resilience.

A new competitor can build faster software. It cannot instantly manufacture decades of reliability, legal relationships, merchant acceptance, and global confidence.

This is why the strongest payment networks should be understood as **trust networks supported by technology**, rather than technology companies that happen to process payments.

Software can be rewritten.

Institutional confidence takes considerably longer.

The Networks Have Survived by Adapting

Visa and Mastercard did not preserve their positions by defending the physical card.

They adapted the network to new payment interfaces.

Magnetic stripes became chips.

Chips became contactless credentials.

Cards moved into digital wallets.

Payment credentials became tokenized.

Fintech companies integrated existing network rails into better consumer experiences.

The visible form of payment changed, while the networks continued operating underneath.

This reveals an important characteristic of durable businesses: they protect the economic function rather than the historical product.

Kodak tried to protect film.

Visa protected authorization, acceptance, routing, security, and settlement.

One of those strategies aged better.

8.2 Evolution of Payment Networks

The development of payment networks can be understood through four major stages.

Stage One — Cash and Local Trust

Early commerce depended heavily on cash and personal relationships.

Transactions were immediate but geographically limited. Cash worked well when buyer and seller were physically present, used the same currency, and trusted the notes being exchanged.

The system became less convenient as commerce expanded.

Cash is simple until it must travel across borders, move through the internet, or be refunded six weeks later.

Stage Two — Bank-Based Payment Instruments

Cheques and bank transfers allowed value to move without physical currency changing hands.

These systems expanded commercial reach but introduced delays, verification requirements, and operational complexity. Payments depended increasingly on financial institutions and clearing arrangements.

The system worked, but not elegantly.

Cheques were effectively handwritten instructions asking a bank to move money later. The financial system accepted this arrangement for an impressively long time.

Stage Three — Global Card Networks

Card networks standardized payment acceptance across merchants and financial institutions.

Visa and Mastercard created shared rules, technical protocols, security processes, and settlement relationships. Consumers could use one credential across a broad merchant base. Merchants could accept payment from customers banking with institutions they had never directly encountered.

The network transformed millions of separate banking relationships into one interoperable commercial system.

This stage created the modern payment-network moat.

Stage Four — Digital and Invisible Payments

The current stage is defined by digital wallets, embedded finance, real-time payments, open banking, tokenization, stablecoins, and artificial intelligence.

The consumer increasingly sees less of the payment process.

A ride-hailing application stores the credential.

A subscription renews automatically.

A smartwatch authorizes a transaction.

A digital wallet replaces the physical card.

Yet underneath the cleaner interface, the same difficult questions remain:

- Who verifies the customer?
- Who routes the transaction?
- Who guarantees settlement?
- Who detects fraud?
- Who handles disputes?
- Who connects institutions across countries?

The future of payments is therefore not the disappearance of networks.

It is the movement of networks deeper into the background.

The payment becomes invisible.

The infrastructure becomes more important.

My Interpretation: From Card Networks to Commerce Networks

Visa and Mastercard are gradually becoming broader **commerce networks**.

Their future is likely to extend beyond traditional card authorization into:

- digital identity;

- cybersecurity;
- data analytics;
- fraud prevention;
- open-banking connectivity;
- account-to-account payments;
- tokenized credentials;
- cross-border settlement;
- business-payment services.

This evolution matters because domestic card payments may face greater fee pressure and alternative payment rails. Expanding into adjacent services allows the networks to capture value from a larger portion of the transaction process.

The companies do not need every future payment to look like a card transaction.

They need to remain useful somewhere in the transaction stack.

8.3 Winners and Losers

The future payment landscape will not produce one universal winner and a collection of extinct competitors. Different organizations will control different layers.

Still, some business models appear better positioned than others.

Likely Winners

Visa and Mastercard

The leading networks remain well positioned because of global acceptance, institutional relationships, scale, security investment, and capital-light economics.

They should continue benefiting from:

- the decline of cash;
- growth in digital commerce;
- international travel;
- emerging-market payment formalization;
- value-added services;
- embedded payments.

Their greatest opportunity is to become neutral infrastructure across multiple payment methods rather than defending cards as the only acceptable model.

Fintech Companies With Distribution or Infrastructure

Fintech companies that solve genuine problems—merchant acceptance, cross-border transfers, software integration, expense management, identity, or fraud—can grow alongside the networks.

The strongest fintech firms will probably not win simply by offering attractive applications. They will need durable distribution, strong unit economics, regulatory capability, and eventually the mildly unfashionable habit of producing cash.

Merchants and Consumers

Greater competition should reduce friction, improve checkout experiences, increase payment choice, and lower costs in selected areas.

Consumers may benefit from faster settlement, better fraud protection, more transparent international transfers, and more convenient financial services.

Merchants may benefit from alternative rails and stronger negotiating leverage.

The payment industry can remain profitable while becoming more efficient. These outcomes are not mutually exclusive.

Cybersecurity and Identity Providers

As commerce becomes increasingly digital, trust becomes more difficult to maintain.

AI-assisted fraud, synthetic identities, account takeovers, and cross-border cybercrime will increase demand for authentication and security services.

Companies capable of identifying legitimate users without turning every purchase into a police interrogation may become increasingly valuable.

Likely Losers

Cash-Dependent Business Models

Cash will not disappear completely, but its share of formal commerce should continue declining over time.

Businesses earning fees from cash handling, manual processing, or inefficient settlement may face structural pressure.

High-Cost Intermediaries With Limited Differentiation

Cross-border payments and bank transfers often involve multiple layers of fees.

Stablecoins, real-time systems, improved banking infrastructure, and fintech competition may compress the economics of intermediaries that add cost without adding meaningful trust, security, or convenience.

Technology is especially dangerous to companies whose main competitive advantage is that customers have not yet found the cheaper button.

Fintech Companies Without Moats

A polished interface is not necessarily a durable business.

Many fintech companies depend on the infrastructure of banks, payment networks, cloud providers, and regulated financial institutions. If they lack proprietary distribution, high switching costs, network effects, data

advantages, or attractive unit economics, their services may become commodities.

The payment industry rewards useful innovation.

It is less generous toward another debit card with a slightly different shade of blue.

Incumbents That Refuse to Adapt

Large financial institutions do not automatically win because of scale.

Banks, networks, and processors that ignore embedded finance, open banking, real-time settlement, AI, or changing merchant preferences could lose valuable customer relationships.

Infrastructure is durable, not immortal.

8.4 Investment Lessons

Studying payment networks provides broader lessons applicable well beyond Visa and Mastercard.

Follow the Value Chain, Not the Product

Investors often focus on the product the consumer sees.

The more valuable position may sit underneath it.

Apple Pay may own the interface.

A bank may own the customer account.

Visa or Mastercard may route the credential.

A processor may serve the merchant.

A cloud provider may host the system.

Several companies can earn money from one transaction.

The investor's task is to determine who captures the most durable economics.

Visibility and profitability are not the same thing.

Small Fees Can Create Enormous Businesses

Visa and Mastercard earn only a small amount relative to the value of each transaction.

That small percentage becomes extremely valuable when applied across trillions of dollars in payment activity.

This model works because the fee is modest enough to remain tolerable while the service is important enough to be difficult to avoid.

Great toll-road businesses do not need to charge every traveller a fortune.

They need an enormous number of travellers.

Network Effects Require Governance

A network is not valuable merely because it is large.

It must be secure, reliable, balanced, and trusted.

Payment networks must manage the interests of:

- consumers;
- merchants;
- banks;
- regulators;
- fintech partners;
- processors.

Charging too much may provoke merchants and regulators.

Weak security may damage consumer trust.

Poor innovation may encourage alternative rails.

Network governance is therefore part of the moat.

The company must strengthen the ecosystem without extracting so much value that participants begin building exits.

A moat can become shallower when the castle owner charges visitors for looking at the water.

Capital-Light Growth Is Especially Powerful

Visa and Mastercard demonstrate why capital efficiency matters.

Their networks can grow without requiring factories, inventory, or direct funding of most consumer credit.

That allows a large portion of operating profit to become free cash flow.

Cash can then be used for:

- technology investment;
- strategic acquisitions;
- dividends;
- share repurchases;
- debt management.

The combination of high returns on capital and a long growth runway is one of the strongest formulas in business.

Growth alone attracts attention.

Capital-efficient growth creates wealth.

Valuation Cannot Be Ignored

Wonderful businesses frequently trade at premium valuations.

That is understandable. Predictable growth, high margins, strong cash flow, and durable moats deserve better valuations than cyclical or low-quality companies.

But quality does not eliminate price risk.

If the market assumes decades of flawless execution, even an excellent business may generate ordinary returns.

The correct question is not:

“Is Visa a great company?”

It clearly is.

The correct investment question is:

“What future growth and profitability are already reflected in the price?”

A wonderful company and a wonderful investment are related concepts.

They are not identical twins.

The Best Risks Are Often Slow-Moving

Visa and Mastercard face real threats:

- regulation;
- domestic payment rails;
- open banking;
- stablecoins;
- merchant pressure;

- cybersecurity;
- government intervention.

However, many of these threats develop gradually.

That gives strong management teams time to respond, acquire capabilities, adjust pricing, and enter adjacent markets.

Businesses are more vulnerable when disruption arrives suddenly and attacks the core product directly.

Payment-network disruption is more likely to occur through gradual fee compression and shifting transaction mix than through one dramatic morning when every merchant on Earth simultaneously deletes Visa.

Investors should still monitor the risks.

They simply should not confuse possibility with immediate probability.

My Payment Network Investment Framework

I evaluate payment companies through five questions.

1. Does the network remain essential?

Would merchants, banks, or consumers experience meaningful disruption if the company disappeared?

2. Is transaction volume expanding?

Does the business benefit from structural growth in electronic commerce, travel, and formal financial activity?

3. Is the company capturing value without weakening the ecosystem?

Are fees sustainable, or are they provoking regulators and customers to build alternatives?

4. Is management adapting to new payment rails?

Can the company participate in open banking, real-time payments, tokenization, stablecoins, and digital identity?

5. Does the valuation permit attractive long-term returns?

How much growth is already priced in, and what happens if the business merely performs well rather than perfectly?

A payment business scoring strongly across all five categories deserves serious consideration.

8.5 Future Outlook

The next decade of payments will likely be shaped by convergence rather than complete replacement.

AI will strengthen fraud prevention while also improving the capabilities of fraudsters.

Stablecoins may reduce friction in cross-border settlement.

CBDCs may modernize sovereign money in selected countries.

Open banking may challenge card economics in specific transaction categories.

Embedded finance will place payments inside software, commerce platforms, vehicles, and connected devices.

Real-time domestic payment rails will continue expanding.

None of these developments automatically eliminates Visa or Mastercard.

The more likely outcome is a multi-rail system.

Consumers and merchants will use different methods depending on cost, convenience, credit, protection, geography, and transaction type. Payment networks will need to connect or support several of those methods.

The future leader may not be the company that owns one exclusive rail.

It may be the company capable of routing commerce across many rails.

Bull Case

The strongest outcome for Visa and Mastercard would involve continued electronic-payment growth, recovering cross-border activity, expanding value-added services, and successful participation in new account-to-account and tokenized-payment systems.

In this scenario, they evolve from card networks into broad global commerce infrastructure.

Base Case

Domestic payment rails and regulation place pressure on selected fees, particularly in lower-risk domestic transactions. However, overall payment volume, cross-border commerce, cybersecurity services, identity products, and digital acceptance continue expanding.

Growth slows modestly, but the economics remain exceptional.

Bear Case

A more difficult outcome would involve:

- widespread low-cost account-to-account merchant payments;
- major regulatory restrictions;
- significant cross-border fee compression;
- reduced bargaining power against large banks and technology platforms;
- failure to participate in tokenized settlement;

- a major cybersecurity or trust failure.

The greatest risk is not any one threat.

It is several threats reinforcing one another while management responds too slowly.

That possibility prevents the investment thesis from becoming automatic.

Even excellent businesses require monitoring.

8.6 Final Conclusion

Visa and Mastercard built extraordinary businesses because they transformed fragmented financial relationships into global commercial infrastructure.

They did not need to issue every card.

They did not need to lend every dollar.

They did not need to own every merchant relationship.

They built the network connecting everyone else.

Their economic strength comes from the interaction of several advantages:

- network effects;
- global scale;
- institutional trust;
- high switching costs;
- standardized infrastructure;

- capital-light growth;
- disciplined capital allocation.

Few companies possess all of these simultaneously.

The most important conclusion from this project is that the payment industry should not be understood as a battle between old cards and new technology.

The plastic card is merely one interface.

The real business lies in establishing trust, routing value, authenticating participants, preventing fraud, creating standards, and connecting institutions.

Those needs will survive whichever device, wallet, token, biometric scan, or mildly unnecessary smart appliance initiates the next transaction.

Visa and Mastercard will not remain dominant simply because their past has been successful. They must continue investing, adapting, and managing the ecosystem fairly enough that participants prefer improvement over replacement.

So far, they have done that remarkably well.

Their greatest competitive advantage may not be any single technology.

It may be their ability to make every new technology compatible with the network they already own.

That is why payment networks deserve to be studied as more than financial companies. They are examples of how an asset-light business, protected by trust and network effects, can scale across the world while collecting a small share of an enormous economic flow.

Or, put more simply:

They built a system where banks take the credit risk, merchants sell the products, consumers make the purchases, and Visa and Mastercard quietly earn a fee for ensuring everyone can find one another.

Business models do occasionally get more exciting.

They rarely get cleaner.